

**PROFESSIONAL
CERTIFICATE
IN MACHINE LEARNING
AND
ARTIFICIAL INTELLIGENCE**

**Office Hour #9 with
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November 10th, 2025 at 7 pm
UTC

Overfitting

[Overfitting](#) refers to a model that models the training data too well.

Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data.

This means that the noise or random fluctuations in the training data is picked up and learned as concepts by the model. The problem is that these concepts do not apply to new data and negatively impact the models ability to generalize.

Overfitting is more likely with nonparametric and nonlinear models that have more flexibility when learning a target function. As such, many nonparametric machine learning algorithms also include parameters or techniques to limit and constrain how much detail the model learns.

Prevent Overfitting

Get More Training Data

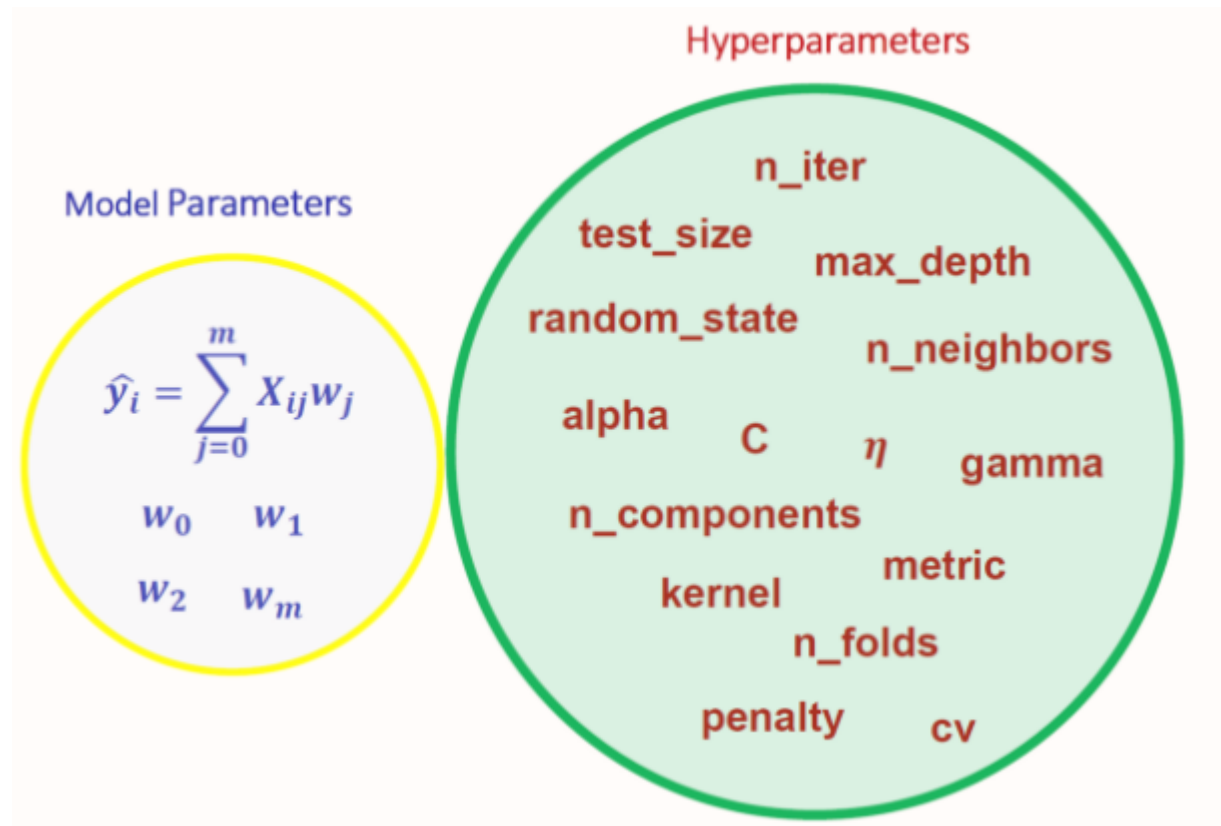
Cross-Validation & HoldOut: Use your initial training data to generate multiple mini train-test splits. Use these splits to tune your model

Augmentation: If you can't get more data, you can try augmentation to add variation in your data. Augmentation means artificially modifying your existing data by means of transforms that resemble the variation you might expect in the real data.

Feature Selection: it is a process by which you automatically search for the best subset of attributes in your dataset.



Automated Hyperparameter Selection



Hyperparameters: adjustable parameters that must be tuned in order to obtain a model with optimal performance (they can be also polynomial)

Sequential Feature Selection

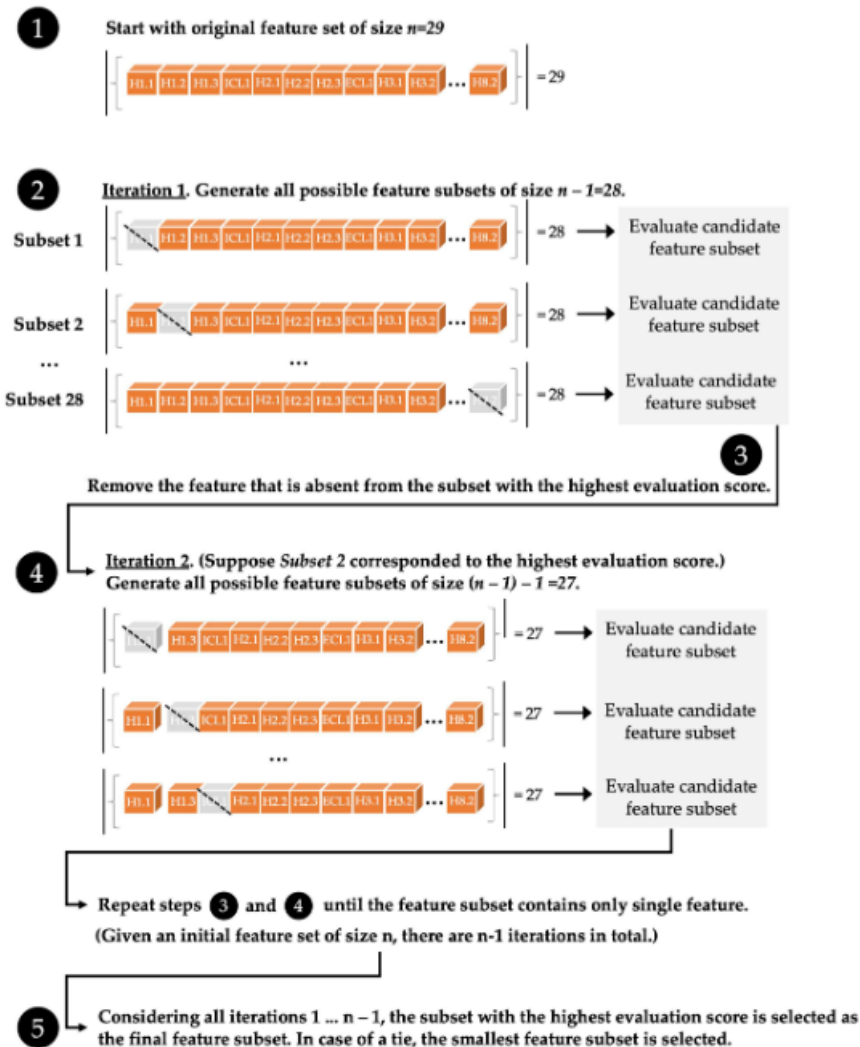
Sequential feature selection algorithms are algorithms that are used to reduce an initial d -dimensional feature space to a k -dimensional feature subspace where $k < d$. The motivation behind feature selection algorithms is to **automatically select a subset of features that is most relevant to the problem**. The goal of feature selection is two-fold: We want to improve the computational efficiency and reduce the generalization error of the model by removing irrelevant features or noise.

In a nutshell, SFAs remove or add one feature at the time based on the classifier performance until a feature subset of the desired size k is reached. There are 4 different flavours of SFAs available:

1. Sequential Forward Selection (SFS) -> features are sequentially added to an empty candidate set until the addition of further features does not decrease the criterion.
2. Sequential Backward Selection (SBS) -> features are sequentially removed from a full candidate set until the removal of further features increase the criterion.
3. Sequential Forward Floating Selection (SFFS)
4. Sequential Backward Floating Selection (SBFS)

The **floating** variants, SFFS and SBFS, can be considered as extensions to the simpler SFS and SBS algorithms. The floating algorithms have an additional exclusion or inclusion step to remove features once they were included (or excluded), so that a larger number of feature subset combinations can be sampled.

Sequential Backward Selection



Regularization

Overfitting of the model occurs when the model learns just ‘too-well’ on the train data. When a model is overtrained on training data, it performs worst on the test data or any new data provided. Technically, the model learns the details as well as the noise of the train data resulting in an hindered performance.

There are several ways of avoiding the overfitting of the model: one of the ways is to apply **regularization** to the model. Regularization is a better technique than Reducing the number of features to overcome the overfitting problem as in regularization we do not discard the features of the model.

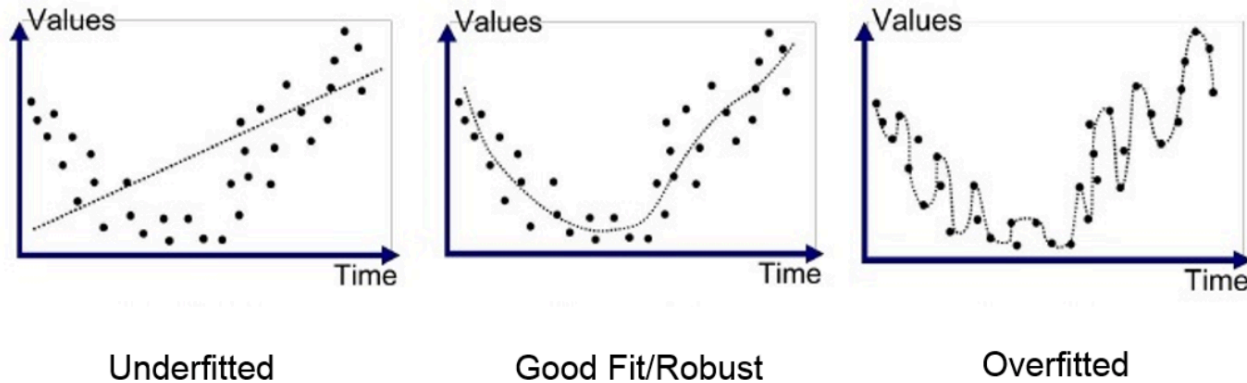
Regularization is a technique that penalizes the coefficient. In an overfit model, the coefficients are generally inflated. Thus, regularization adds penalties to the parameters and avoids them weigh heavily. The coefficients are added to the cost function of the linear equation. Thus, if the coefficient inflates, the cost function will increase.

A simple relation for linear regression looks like this.

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

The fitting procedure involves a loss function, known as residual sum of squares or RSS. The coefficients are chosen, such that they minimize this loss function.

$$\text{RSS} = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$



Underfitting VS Good Fit(Generalized) VS Overfitting

Underfitting occurs when the model doesn't work well with both training data and testing data (meaning the accuracy of both training & testing datasets is below 50%)

A model is a **Good Fit** when it works well with both training and testing datasets (meaning the accuracy for both datasets is around 70%–85% in general cases).

Overfitting occurs when the model works very well with the training dataset, but on the testing dataset, it fails (meaning that the accuracy of training data is >90% and testing data is < 65%).

Ridge Regression

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2$$

RSS, measures the amount of error staying between the regression function and the data set. It is a measure of the amount of statistical variance present in the dataset.

In Ridge regression, RSS is modified by adding the shrinkage quantity. λ is the tuning parameter that decides how much we want to penalize the flexibility of our model. The increase in flexibility of a model is represented by increase in its coefficients, and if we want to minimize the above function, then these coefficients need to be small. This is how the Ridge regression technique prevents coefficients from rising too high. Naturally, selecting a good value of λ is critical.

Lasso Regression

Least **Absolute Shrinkage and Selection Operator**.

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|.$$

***This variation differs from ridge regression only in penalizing the high coefficients.** It uses $|\beta_j|$ (modulus) instead of squares of β , as its penalty.*

Thus, from watching the cost function, the only **difference** found between **Ridge & Lasso** is that: *instead of taking the square of the coefficients, magnitudes are taken into account.*

This type of regularization can lead to zero coefficients, i.e. some of the features are completely neglected for evaluating output. **So Lasso regression not only helps in reducing overfitting but can help us in feature selection.** Ridge regression only reduces the coefficients close to zero but not zero, whereas Lasso regression can reduce coefficients of some features to zero, thus resulting in better feature selection. Same as in regression, where also the hyperparameter Lambda can be controlled and all the other functioning works the same here.

QUESTIONS?

